

Mukesh Patel School of Technology Management & Engineering
School of Technology Management and Engineering

Course Policy Document

Course Name - (Code): Vulnerability Assessment and Penetration Testing – (702IT0C018)

Program and Semester: B. Tech Computer Science and Engineering (Cybersecurity), Sem VI	Pre-requisite Course: Cybersecurity Fundamentals, OS and Computer Networks					
Academic Year: 2025-26	Credit Details:	L	T	P	C	H
		2	0	2	3	4
Name of Course Faculty: Prof. Pintu Shah Program: B Tech(CSE-Cybersecurity), Div K, Mumbai Campus Contact Details: pintu.shah@nmims.edu Office Hours:	Faculty associated with the course: - 1. 2.					
Pre-Course Activity: There are no pre course activity						
Course link: https://portal.svkm.ac.in/usermgmt/login Course code to join class MS Team: dq8w5sm						

1. Introduction to the Course

1.1 Importance of the Course

1.1.1 Domain Relevance:

Cybersecurity testing is essential for identifying vulnerabilities and weaknesses in an organization's IT infrastructure. By conducting thorough testing, such as vulnerability assessments, penetration testing, and security audits, businesses can uncover potential security flaws before they are exploited by malicious actors. This proactive approach helps protect sensitive data, intellectual property, and customer trust. Cybersecurity testing also ensures compliance with regulatory requirements and industry standards, mitigating legal and financial risks. As cyber threats continue to evolve, regular testing allows organizations to adapt to emerging risks and stay one step ahead of attackers. Ultimately, cybersecurity testing strengthens an organization's overall security posture, preventing costly breaches, minimizing downtime, and maintaining business continuity.

1.1.2 Industry Relevance:

Vulnerability Assessment and Penetration Testing (VAPT) is highly relevant across industries due to the increasing prevalence of cyber threats. For sectors like finance, healthcare, and e-commerce, where sensitive data is crucial, VAPT helps identify vulnerabilities that could lead to data breaches or regulatory violations. In industries like government and defense, VAPT ensures national security and protects critical infrastructure from cyberattacks. The rise in digital transformation means businesses in all sectors face new risks, making regular VAPT a vital part of their cybersecurity strategy. Additionally, compliance with industry standards such as GDPR, HIPAA, and PCI-DSS often requires VAPT. By detecting vulnerabilities, VAPT strengthens an organization's defenses, ensuring business continuity and maintaining trust with customers and stakeholders.

1.2 Objectives of the Course:

- 1.2.1** To simulate various attacks in the lab environment using the tools used by attackers in real life.
- 1.2.2** To suggest suitable counter measures for the vulnerabilities found during the VAPT.

2. Course Outcomes (CO), Mapping with Program Outcomes (PO), and Program Specific Outcomes (PSO)

2.1 Course Outcomes

- 2.1.1 CO1:** Demonstrate hacking in a lab environment
- 2.1.2 CO2:** Describe various countermeasures
- 2.1.3 CO3:** Describe various professional, ethical and legal issues related to ethical hacking

2.2 Program Outcomes(PO) the course contributed to:

- 2.2.1 PO-1 - Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization to develop to the solution of complex engineering problems.
- 2.2.2 PO-2- Problem analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.
- 2.2.3 PO-3 - Design and development of solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes

to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required.

- 2.2.4 PO-4 - Conduct investigations of complex problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions.
- 2.2.5 PO-5 - Modern tool usage:** Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems.
- 2.2.6 PO-6 - The engineer and society:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.
- 2.2.7 PO-7 - Ethics:** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws.
- 2.2.8 PO-8 - Individual and Collaborative Team work:** Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
- 2.2.9 PO-9 - Communication:** Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.
- 2.2.10 PO-10 - Project Management and Finance:** Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
- 2.2.11 PO-11 - Life-Long Learning:** Recognize the need for, and have the preparation and ability for independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change.

2.3 Program Specific Outcomes (PSO):

- 2.3.1 PSO-1:** Demonstrate an ability to analyze and mitigate emerging cyber threats by applying cybersecurity principles and practices.

- 2.3.2 PSO-2:** Demonstrate an ability to recommend appropriate cybersecurity solutions considering business goals, industry standards and regulatory compliance requirements.

CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	3	2	2	3	–	1	2	1	–	2	3	2
CO2	2	3	3	2	2	2	1	1	1	–	2	3	3
CO3	1	1	–	–	–	3	3	1	2	1	2	2	2

Mapping Levels: 3- High, 2-Medium, 1-Low

2.4 Student Outcomes (SO) (For ABET accredited Programs):

- 2.4.1 SO-1:** Analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions.
- 2.4.2 SO-2:** Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.
- 2.4.3 SO-3:** Communicate effectively in a variety of professional contexts.
- 2.4.4 SO-4:** Recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.
- 2.4.5 SO-5:** Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.
- 2.4.6 SO-6:** Apply security principles and practices to maintain operations in the presence of risks and threats.
- 2.4.7 SO-7:** An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.
- 2.4.8 SO-8:** An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

3. Teaching-learning methodology

This course will be taught using problem and project based learning methodology. Students are expected to carry laptop for the class. Each group will be given a vulnerable VM in the first week of the semester. They will create lab environment on their laptop using virtual box and following software

- Kali Linux
- Parrot OS
- Vulnerable VM
- Virtual Box

3.1 Instruction Plan

Lecture No.	Topic	Teaching Method *	Blooms Level	Resources	COs mapped	Assessment and Evaluation
1	Introduction to the course	Presentation	Understand			
2	Need for adversarial thinking and penetration testing		Understand	TB	1	CT1, TEE
3	Ethics of hacking, hacking process		Understand	TB	1	CT1, TEE
4	Types of hackers, types of penetration testing, red/blue teaming		Understand	TB	1	CT1, TEE
5	Testing methodologies (OSSTMM)		Understand	TB	1	CT1, TEE
6	Testing methodologies (PTES)		Understand	TB	1	CT1, TEE
7	Testing methodologies (OWASP Testing Guide)	PPT, Whiteboard	Understand	TB	2	CT1, TEE
8	Rules of engagement, Vulnerability metrics (CVE, CWE, CVSS).		Understand	TB	2	CT1, TEE
9	Reconnaissance and scanning Introduction, types of reconnaissance		Understand	TB	2	CT1, Lab and Project, TEE
10	Various techniques of recon (social engineering, web based recon, DNS based recon, network based recon, Google hacking etc.) and countermeasures		Apply	TB	2	CT1, Lab and Project, TEE
11	Various techniques of recon (social engineering, web based recon, DNS based recon, network based recon, Google hacking etc.) and countermeasures (Contd.)		Apply	TB	2	CT1, Lab and Project, TEE

12	Test 1					
13	Various techniques of recon (social engineering, web based recon, DNS based recon, network based recon, Google hacking etc.) and countermeasures (Contd.)		Apply	TB	2	Lab and Project, TEE
14	Various techniques of recon (social engineering, web based recon, DNS based recon, network based recon, Google hacking etc.) and countermeasures (Contd.)		Apply	TB	2	Lab and Project, TEE
15	Port scanning		Apply	TB	2	CT2, Lab and Project, TEE
16	Network scanning	PPT, Whiteboard	Apply	TB	1	CT2, Lab and Project, TEE
17	Vulnerability scanning and Sniffers.		Apply	TB	1	CT2, Lab and Project, TEE
18	Exploitation Password cracking		Apply	TB	1	CT2, Lab and Project, TEE
19	Spoofing, Session hijacking, DoS / DDoS,		Apply	TB	1	
20	Buffer Overflow, malware, evading firewall and IDS		Apply	TB	1	CT2, Lab and Project, TEE
21	SQL Injection,	PPT, Whiteboard	Apply	TB	3	CT2, Lab and Project, TEE
22	SQL Injection (Contd.)		Apply	TB	3	CT2, Lab and Project, TEE

23	OWASP top 10 web application vulnerabilities		Apply	TB	3	CT2, Lab and Project, TEE
24	Test 2					
25	OWASP top 10 web application vulnerabilities (Contd.)		Apply	TB	3	CT2, Lab and Project, TEE
26	Hacking wireless networks	PPT, Whiteboard, YT Videos	Apply	TB	3	Lab and Project, TEE
27	AV evasion, privilege escalation.		Apply	TB	3	Project, TEE
28	Legal, Professional and Ethical issues Cyber laws, various ethical dilemma, professional conduct,		Understand		3	TEE
29	Responsible disclosure and penetration testing report writing		Understand	TB		Project
30	Course Wrap-up					

4. Assessment and Evaluation Scheme

	Internal Continuous Assessment (ICA) (50 Marks)				
	Class Tests (20 Marks)		Term-work (30 Marks)		
Assessment Component	Class Test-1	Class Test-2	Lab Submissions	Project / Case study	Class Participation
Marks	10	10	10	10	10

4.1 Internal Continuous Assessment (ICA) – 50 marks

1. Class test 1 and 2

- Two class tests will be conducted as per the academic calendar.
- It will be lab exam.
- It will be conducted offline for 10 marks each.

2. Lab performance evaluation (10 marks)

Each lab will be assessed for 10 marks. Following points will be considered while assessment

- a. Lab implementation and troubleshooting (6 marks)
- b. Documentation and timely submission (4 marks)

Note: **50% penalty for late submission.**

3. Group Project (10 marks)

- a. CR will form a group of 2-3 students. This activity should be completed in the first week.
- b. Each group will be assigned one vulnerable machine.
- c. Each group will hack the given vulnerable machine
- d. Report and presentation submission deadline is March **31, 2026**.

4. **Class Participation / Cisco certification (10 marks)** – Each student will be enrolled in the ethical hacking course offered by Cisco Academy. Each student is expected to complete the course along with all activities described in the course. This will be counted towards class participation along with other in-class activities.

4.2 Term End Examination (TEE) – (100 marks scaled down to 50)

Final lab TEE will be conducted at the end of the semester. It will be for 100 marks (3-hour duration). Students will be given vulnerable VM. They will be required to hack the VM and answer review questions.

4.3 Course Passing Criteria

4.3.1 ICA (50 marks) – No minimum marks

4.3.2 TEE (100 marks scaled to 50) – 40% required for passing

4.4 Assessments and Mapping to Course Outcomes

	Internal Continuous Assessment (ICA)					Term End Examination (TEE)
Course Outcomes	CT-1	CT-2	Lab Submissions	Project	Cisco Certification	TEE
CO-1	Y	Y	Y	Y	Y	Y
CO-2	Y	Y	Y	Y	Y	Y
CO-3	Y		Y	Y	Y	Y

5. Laboratory details

Tentative list of the lab experiment is given below. Faculty may revise this list during the semester.

Exp. No.	Week No.#	Programming Topic	Mapped CO
1.	Week 2	lab 1: Reconnaissance and foot printing	1&2
2.	Week 3	Lab 2: Vulnerability Scanning	1&2
3.	Week 4	Lab 3: Password Cracking	1&2
4.	Week 5	Lab 4: Exploitation	1&2
	Week 6	Revision	
5.	Week 7-8	Lab 5: Social Engineering	1&2
6.	Week 9-10	Lab 6: CFT (Scanning and Enumeration)	1&2
7.	Week 11-12	Lab 7: CFT (Exploitation and Privilege Escalation)	1&2
8.	Week 13	Lab 8: CFT (Report Writing)	1,2,&3
9.	Week 14	Lab 9: Equifax Data Breach Case Study	1,2,&3
	Week 15	Revision	

6. Tutorial Plan

This course does not have any tutorial.

7. Course Material

References and Lab Manuals would be uploaded on LMS by faculty every week.

8. GenAI Usage

8.1 Group Project:

- Students are permitted to use GenAI for exploring countermeasures and ideas related to exploit.

9. Academic Integrity Statement

Faculty expects academic integrity from all students. Student should refer to student resource book (SRB) for academic guidelines. Original work expected from students for all of the **assigned assessment work**. Copying in any form not acceptable and will invite disciplinary action. Evaluation of corresponding component will be affected proportionately in such cases. Plagiarism detection software will be used to check plagiarism wherever applicable.

*** - Only Teaching Method in the Instruction Plan for the course may vary for different faculty teaching the course, rest of the Course Policy Document will not change.**